TraceMemory: allocation, correction magnitude, and source memory

Fixed source reference

Source image → controlled inversion → $r_{0:T}$; $p_i = (r_{i+1} - r_i)/h_i$

Request-conditioned protection

Three paired native source/target probes

Channel RMS → percentile clipping → E

$P = 1 - 0.75E \in [0.25, 1]$

Lower prompt response gives stronger correction.

Residual at the current edited state

$c_i = p_i - u_i^s(z_i)$

Reference motion uses r_i; source field uses z_i.

Each branch evaluates its own evolving state.

Source-field evaluation uses native attention.

Three levels of correction control

1 Native operating points

Fixed nominal coefficient

CORR25: $0.25P \odot c_i$

U25: $0.25c_i$

Applied RMS depends on weights and current residual.

2 Common-state RMS

CORR25 / ENERGY

$$\hat{c}_i^{\text{ENERGY}} = 0.25\frac{\text{RMS}(P \odot c_i)}{\text{RMS}(c_i)} c_i$$

Equal RMS at a common state; recomputed on each rollout.

Realized doses can diverge.

3 Actual-step RMS

Shared anchor profile $b_{s,i}$; fixed β

Request	Global	Shift	PERM
$P \odot c_i$	c_i	$\widetilde{P} \odot c_i$	$P^{\pi} \odot c_i$

$$\hat{c}_i = \beta b_{s,i} v_i / \text{RMS}(v_i)$$

$$\text{RMS}(\hat{c}_i) = \beta b_{s,i} \quad (v_i \neq 0)$$

Separate source-memory intervention

Source K/V captured at r_i; full source pool.

CORR25: $M_q = 1$; BOTH25: pooled P.

Memory active at steps 0-6, then native attention.

Actual-step direction branches keep memory fixed.

Target field acts over the full image

$$z_{i+1} = z_i + h_i[\, u_i^{t,M}(z_i) + \hat{c}_i \,]$$

Direction branches share the same time nodes.

Final state z_T → VAE decode → edited image.